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AMX Switchboards

CATALOG



AMX LOW-VOLTAGE SWITCHBOARD SOLUTIONS

Why AMX?

AMX switchboards are bolted modular, low-voltage power distribution assemblies, fully type-tested to UL891 designed to deliver safety, uptime, and serviceability in critical environments.

High uptime

AMX switchboards are designed for continuous operation using a breaker-centric protection architecture. Integration of UL1066 ACBs and UL489 MCCBs, combined with a fully rated bus system and short-time withstand capability up to 100 kA for 1 second, ensures predictable fault behavior. Selective coordination limits outages to affected circuits, keeping critical loads energized and reducing recovery time.

Compact density

High current ratings and optimized breaker stacking allow AMX to deliver dense power distribution in a compact footprint. Busbars rated from 1200 A to 6000 A, standardized section widths, and front access to instrumentation maximize feeder capacity while minimizing electrical room space and installation constraints.

Proven performance

AMX switchboards are type tested as complete assemblies to UL 891, verifying thermal performance, dielectric integrity, and short-circuit withstand. Seismic-qualified configurations are available where required. This testing confirms real-world performance and reduces commissioning and operational risk.

Design flexibility

The AMX platform supports modular standardized construction. Design configurations, breaker- and meter-agnostic integration, and support for Modbus TCP/IP and Modbus RTU enable alignment with customer standards and seamless integration into monitoring and control systems.

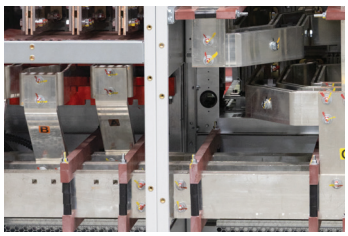
Digital ready

AMX uses high accuracy metering that seamlessly integrate with the data center's BMS for remote, real time power quality metrics. HMIs and pilot device lights also provide local indications for confident operations

Data center ready

Rated up to 600 V AC with natural convection cooling and indoor NEMA 1 construction, AMX is purpose-built for data centers and critical facilities. Front-access operation, scalable sections, and robust protection features support long-term reliability, operational efficiency, and future expansion.

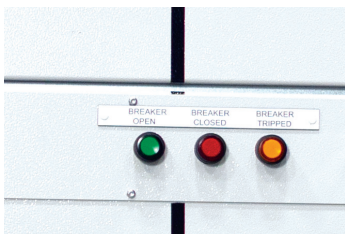
Benefits of AMX switchboards

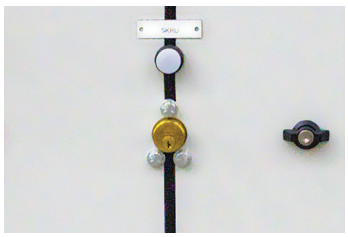












Construction and design

- The AMX switchboard uses a modular, bolted design to support scalability and future expansion
- Rear-isolated, tin-plated copper busbars provide reliable current distribution up to 6000 A
- Compact enclosure minimizes electrical room space requirements
- Optimized breaker stacking maximizes feeder density within each section

Safety and protection

- Dead-front construction prevents contact with live parts during operation and maintenance
- Segregated ACB compartments limit fault propagation and enhance personnel safety
- Guarded MCCB terminals provide safe isolation to energized connections
- Breaker-based selective coordination isolates faults without unnecessary outages



Operation and maintenance

- Clear ON, OFF, and TRIP indication improves operator awareness and response
- LOTO-ready breaker mechanisms support safe maintenance procedures
- Mechanical and kirk key interlocks prevent unsafe operating sequences

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Benefits of AMX switchboards

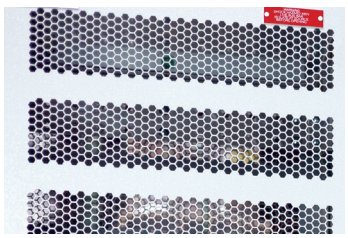






Digital readiness and monitoring

- Meter-agnostic architecture supports flexible monitoring configurations
- Modbus TCP/IP and Modbus RTU enable integration with BMS and EMS platforms
- Customer-defined metering points provide targeted system visibility
- Front-accessible instrumentation allows safe interaction in live environments



Reliability and uptime

- A fully rated system ensures predictable performance without derating
- High interrupting breakers provide fast and reliable fault clearing
- Natural convection cooling reduces system complexity and maintenance needs
- The design supports preventative maintenance required for mission-critical environments



Certification and testing

- The switchboard is tested to UL 891 as a complete assembly
- Busbars are verified for short-time withstand up to 100 kA
- OSHPD-compliant seismic options are available for regulated installations
- Manufacturing follows an ISO 9001–certified systems

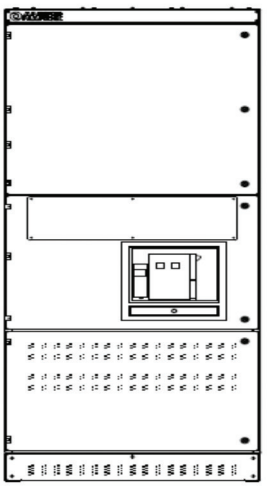
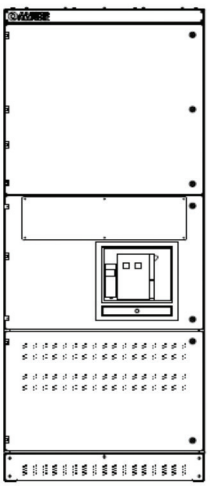
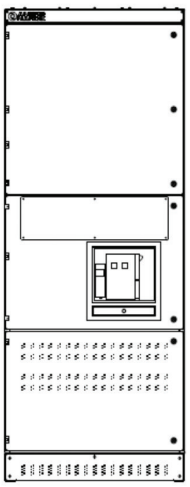
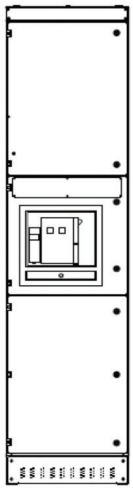
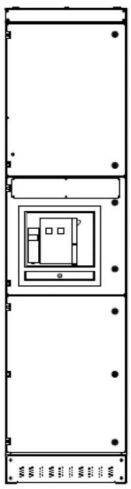
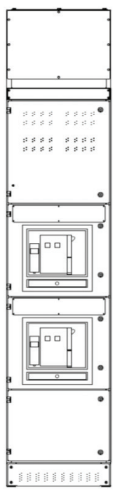
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Technical specifications

Parameter	Specification
Product	AMX low voltage switchboards
Applicable standard	UL 891
Seismic qualification	OSHPD compliant to ICC-ES AC156
Markets	Data centers, AI/HPC loads, cloud/colocation facilities, critical infrastructure
Installation type	Indoor, isolated rear through bus, freestanding
Electrical	
Rated operational voltage	Up to 600V
Rated frequency	60Hz
Horizontal busbar rating	1200A to 6000A (density rated)
Short time withstand	Up to 100kA
Neutral bus	100%
ACB configuration	Fixed or withdrawable
MCCB configuration	Fixed (group mounted)
Construction	
Construction type	Bolted, modular
Enclosure material	Galvanized steel
Enclosure rating	NEMA 1 (indoor)
Busbar material	Tin plated copper
Insulation system	SMC bus supports, GPO3 insulating screens
ACB stacking	Up to 2 ACBs per column
MCCB stacking	250A: up to 16 600A: up to 11 800 -1200A: up to 6
Safety	
Personnel protection	Dead front construction, touch safe instrumentation
Interlocking	Mechanical interlocks, kirk key systems
Breaker safety	Shutters and mechanical interlocks (where applicable)
Environment	
Operating temperature	-5°C to +40°C (maximum +40°C)
Relative humidity	30% at 40°C to 90% at 20°C
Cooling method	Natural convection
Controls and communications	
Communication protocols	Modbus TCP/IP, modbus RTU
System integration	BMS / EMS / monitoring systems (per customer specification)

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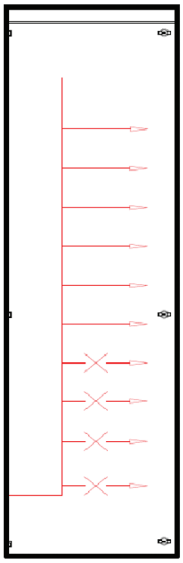
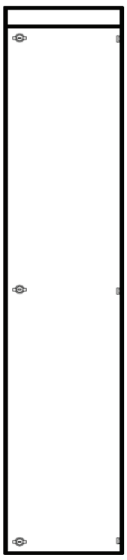
Standard cabinet configurations

6000A Single Stack	**5000A/4000A** Single Stack	**3200A/3000A** Single Stack	**2500A** Single Stack	**2000A-1000A** Single Stack	**1600A-1000A** Double Stack
					

Cabinet type	**3Ph/3W width**	**3P/4W width**	**Depth**	**Height**	**# of devices**	**Voltage**	**Device type**	**Main bus rating**	**Cable entry/ exit**
6000A	45”	48”	36”	91.5”	Single ACB	Up to 600V	Fixed or drawout	Density rated	Top or bottom
5000A/4000A	36”	40”	36”	91.5”					
3200A/3000A	27”	32”	32” or 36”	91.5”					
2500A	27”	27”	32” or 36”	91.5”					
2000A-1000A	22”	22”	32” or 36”	91.5”					
1600A-1000A	22”	32”	32” or 36”	91.5”	Double ACB				Top

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Standard cabinet configurations

MCCB stack	Cable way	Marshalling	Transition
			

Cabinet type	3Ph/3W width	3P/4W width	Depth	Height	Voltage	Device type	# of devices	Cable entry / exit
MCCB stack	24”	24”	32 or 36”	91.5”	Up to 600V	Fixed (group mounted)	250A: up to 16 600A: up to 11 800-1200A: up to 6	Top or bottom
Cable way	16” or 32”	16” or 32”	32” or 36”	91.5”	-	-	-	-
Marshalling	16” or 32”	16” or 32”	32” or 36”	91.5”	-	-	-	-
Transition	16”	16”	32” or 36”	91.5”	-	-	-	-

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Services and lifecycle support

Anord Mardix provides structured services to support AMX UL891 switchboards throughout the project lifecycle. Services are delivered by experienced field and engineering teams to ensure correct installation, verified performance, and smooth handover to operations.

1. Project and pre installation support

Includes technical review against project one-line diagrams and site requirements, installation guidance for layout, clearances, cable entry, and environmental conditions, and provision of drawings, datasheets, and test records for submittal and site use.

2. Testing and verification

Includes verification of protection settings against the approved coordination study, operational checks of withdrawable breakers, interlocks, and safety features, and validation of Modbus TCP/IP and Modbus RTU communications where provided.

3. Commissioning and startup

On-site startup support is provided to verify installation readiness for energization, followed by functional testing of breakers, interlocks, instrumentation, and control circuits, and integrated testing with upstream and downstream systems prior to handover.

4. Training and handover

Training covers normal operation, breaker handling, safety features, isolation procedures, racking operations, and inspection practices, supported by a review of handover documentation for ongoing operation.

5. After sales support

After commissioning, support includes technical assistance for operational questions, guidance on spare parts, and field service support for troubleshooting or modifications as required.

6. Warranty

A standard 12-month defects warranty is provided. Extended warranty options are available subject to scope and agreement.

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Appendix

Application considerations and definitions

ACB (Air Circuit Breaker)

A high-capacity power circuit breaker is integrated into the AMX switchboard as an incoming, tie, or large feeder device. In AMX, ACBs are provided in fixed or withdrawable configurations, installed in segregated compartments, and engineered to support system-wide selectivity, short-time withstand, and safe maintenance isolation.

AMX switchboard

A modular, low-voltage power distribution assembly type-tested to UL 891. AMX switchboards are designed for high-density, mission-critical applications and integrate ACBs, MCCBs, busbars, protection, and control systems into a scalable, front-access enclosure.

Breaker-agnostic design

An AMX design philosophy that allows integration of approved breakers from multiple manufacturers without changing the core switchboard architecture. This enables flexibility in protection strategy, supply chain resilience, and customer standardization.

Breaker-based protection

A protection architecture in which fault interruption, isolation, and coordination are achieved primarily through the characteristics and settings of ACBs and MCCBs integrated into the AMX switchboard.

Busbar (horizontal / through-bus)

The primary current-carrying copper conductors in the AMX switchboard that distribute power across sections. AMX busbars are tin-plated copper, rear-isolated, mechanically braced, and rated from 800 A up to 6000 A to support high-density power distribution.

Closed-door operation

An AMX safety feature that allows breaker operation and status verification with compartment doors closed, maintaining the integrity of dead-front protection and reducing operator exposure during energized operation.

Dead-front construction

A fundamental AMX safety design in which live electrical parts are fully enclosed behind barriers and covers. This ensures operators cannot contact energized components during normal operation or routine maintenance.

Drawout (withdrawable) ACB

A configuration in which the ACB is mounted on a racking mechanism within the AMX switchboard, allowing it to be moved between connected, test, and disconnected positions. This enables safe isolation, inspection, and maintenance without disturbing upstream busbars.

Front access serviceable

An AMX layout feature in which all normal operation, metering, breaker servicing, and preventative maintenance activities are performed from the front of the switchboard.

Fully rated system

An AMX design approach where all breakers and busbars are applied at their full continuous current rating without derating or reliance on diversity factors, ensuring predictable thermal and protection performance.

Ingress protection (NEMA 1)

The enclosure rating of the AMX switchboard, intended for indoor installations. NEMA 1 construction protects against accidental contact with live parts and falling debris but is not intended for outdoor or wet environments.

Interlocking (kirk key)

A mechanical key-based interlocking system integrated into AMX switchboards to enforce safe operating sequences, such as preventing simultaneous connection of multiple sources or access to compartments under unsafe conditions.

Appendix

Application considerations and definitions

LOTO-ready isolation

An AMX safety feature that provides lockout/tagout capability at breaker operating mechanisms, allowing maintenance personnel to secure isolation points in compliance with site safety procedures.

MCCB (Molded-Case Circuit Breaker)

A compact UL 489 circuit breaker used in the AMX switchboard for feeder and branch circuit protection. MCCBs are fixed in a group mounted and are arranged to maximize feeder density while maintaining guarded terminations and dead-front safety.

Rear-isolated busbars

An AMX busbar arrangement where high-energy conductors are located at the rear of the enclosure and separated from operator-accessible areas. This improves personnel safety and allows safe access to instrumentation and control components.

Selective coordination

A protection strategy implemented within the AMX switchboard whereby breaker time-current characteristics are selected so that only the breaker closest to a fault operates, maintaining power to unaffected sections

Short-circuit withstand rating

The verified ability of the AMX switchboard, including busbars and supports, to safely withstand and carry high fault currents (up to 100 kA) without mechanical or thermal damage.

Stacking density

A measure of how many breakers can be installed vertically within an AMX section. AMX designs optimize stacking density to deliver high feeder counts while maintaining thermal performance and service clearances.

Switchboard (UL 891)

The UL classification of the AMX assembly, defining it as a low-voltage power distribution unit with front-accessible overcurrent devices, and instrumentation, evaluated to UL 891 construction and test requirements.

Thermal type testing

Factory validation performed on AMX switchboards to confirm that temperature rise of busbars, breakers, and terminals remains within allowable limits at rated current under worst-case conditions.

Touch-safe instrumentation

Instrumentation and control devices in the AMX switchboard that are installed behind protective barriers or covers, allowing interaction without exposure to energized components.

Type tested assembly

A switchboard that has been tested as a complete system (not just individual components) for thermal performance, dielectric integrity, and short-circuit withstand in accordance with UL891 standard.

Withdrawable maintenance isolation

The ability, provided by drawout ACBs in AMX switchboards, to physically isolate a breaker from live busbars while maintaining enclosure integrity, enabling safer maintenance in live facilities

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Anord Mardix, a Flex company, is a leading provider of power distribution and protection equipment to the global market. We are a vendor of choice to many of the world's core markets, from data centers to cloud computing.

With over 100 years of combined expertise we offer clients the most comprehensive product range in the industry. Our ability to customize our product offering to each project, and manufacture on a global scale, ensures reliable and bespoke end-to-end critical power solutions to all our customers, from independent providers to hyperscale leaders.

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